

Brain Tumor Methylation Reference Service

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CLIA 14D9928844
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DNA METHYLATION-BASED CNS TUMOR CLASSIFIER (v12.5) — FINAL REPORT

PATIENT MARTINEZ-LANG, ELENA J. MRN MRN-3318477 47 y / Female	SPECIMEN Reference accession BTMRS-26-M-0907 Client SP-26-N-04412 FFPE, ~24h fixation DNA extracted 2026-04-23	ORDERING Northshore University Hospital Reyes, M., MD Received 2026-04-20 Reported 2026-04-29
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CLASSIFIER RESULT

Methylation family: Diffuse glioma, IDH-mutant
Methylation class: Astrocytoma, IDH-mutant (high-grade)
Calibrated score: 0.97 (threshold for classification: ≥ 0.90)
MGMT promoter status (by array beta-value model): Unmethylated

METHODOLOGY

Genomic DNA was extracted from FFPE tumor tissue (cassette A4). DNA (500 ng) was bisulfite-converted and assayed on the Illumina Infinium MethylationEPIC v2.0 array (~935,000 CpGs). Raw IDAT files were processed by an in-house pipeline (preprocessNoob, BMIQ, batch-effect correction); the resulting beta-value matrix was submitted to the CNS Tumor Methylation Classifier (v12.5, publicly available reference cohort). A random-forest classifier assigns one of 184 methylation classes; calibrated scores derived via Platt scaling on an independent validation cohort.

TOP CLASSIFICATION CANDIDATES (calibrated scores)

Methylation class	Family	Score
Astrocytoma, IDH-mutant (high-grade)	Diffuse glioma, IDH-mutant	0.97
Astrocytoma, IDH-mutant (low-grade)	Diffuse glioma, IDH-mutant	0.02
Oligodendroglioma, IDH-mutant + 1p/19q codeleted	Diffuse glioma, IDH-mutant	0.00
Glioblastoma, IDH-wildtype, methylation class mesenchymal	Glioblastoma, IDH-wildtype	0.00
Reactive tumor microenvironment / control tissue	Control / normal	0.00

The top class receives a calibrated score well above the 0.90 threshold for high-confidence classification. The second-ranked class is the low-grade variant of the same family; this does not alter the entity assignment but is consistent with grading by histology + molecular criteria as reported separately.

COPY-NUMBER PROFILE (FROM ARRAY)

Array-derived copy-number profile is largely balanced. No **CDKN2A/B (9p21.3) homozygous deletion**. No EGFR amplification. No chromosome 7 gain / chromosome 10 loss. No PDGFRA amplification. No focal MYC / MYCN amplification. The CDKN2A/B status is concordant with the panel NGS report (accession HX-NGS-CNS-018244).

INTERPRETATION

The methylation profile classifies as **Astrocytoma, IDH-mutant (high-grade)** with a calibrated score of 0.97 (high confidence). This is concordant with the histologic (diffuse astrocytic morphology) and molecular (IDH1 R132H, ATRX loss, TP53 R175H, 1p/19q intact, CDKN2A/B diploid, MGMT unmethylated) findings. Methylation grading nomenclature recognizes a high-grade vs low-grade split; final WHO grade assignment under CNS5 is derived from integration of histology (mitotic activity, microvascular proliferation, necrosis) and molecular criteria (CDKN2A/B homozygous deletion). The classifier result alone is consistent with grade 3 or 4 but does not establish the grade independently.

LIMITATIONS

The methylation classifier is intended for adjunctive use alongside histopathology and molecular testing. Calibrated scores below 0.90 should not be used for classification. Tumor purity, age effects on the FFPE block, and array-batch variation can influence performance. This report does not assign a WHO grade. The classifier reflects the v12.5 reference cohort and may differ from prior versions.

Electronically signed: Hofmann, Kerstin H., PhD, Director (Neuropathology Bioinformatics), and Adebayo, Tunde O., PhD, Senior Bioinformatics Scientist — 2026-04-29.

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